Illusions of Impunity: Deconfounding Fixture Quality, Non-Foul Intercepts, Spatial Placement, and Small-Sample Noise in Association Football Disciplinary Ratios

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Abstract

Public and media controversies in association football routinely cite a club's ratio of yellow cards to fouls committed as evidence of refereeing bias or tactical impunity. Across a four-study empirical programme combining 58,013 completed-season matches across 11 European top divisions (2000–2025), 2,737 harvested Portuguese referee appointments, 47,955 spatio-temporally coded fouls from five major leagues, and 3,100 historical four-match season openings, we show that raw cards-per-foul ratios are systematically confounded at four distinct levels. First, an audit of 286 league-season files demonstrates that 107 files return valid tables with zero match statistics while referee identifiers are absent in 9 of 11 leagues, necessitating within-match conditioning and external official harvesting. Second, a club's raw booking index is confounded by pre-match win expectation (31.3% underdog-to-favourite swing in Portugal; 1.098 vs. 0.856 median across 11 leagues) and by non-foul card intercepts (13%–47% of yellow cards arise from dissent, time-wasting, and simulation, causing observed yellows-per-foul to fall monotonically from 0.197 at ≤ 6 fouls to 0.122 at > 20 fouls). Crucially, computing the identical index for a club's *opponents* reveals that team strength increases the opponent's booking index ($r = +0.283, p = 6.6 \times 10^{-6}$) and that holding club identity fixed across opponent-strength terciles raises booking rates in 11 of 11 leagues ($0.938 \to 0.993 \to 1.088$): card intensity belongs to fixture quality rather than club favouritism. After leave-one-club-out referee and situation adjustment in Portugal, FC Porto ($0.890 \to 1.002$) and SL Benfica ($0.885 \to 1.002$) sit exactly at expectation, while only Sporting CP (1.177, Benjamini–Hochberg $p = 0.0004$) separates among 26 clubs. Third, at the individual foul level ($N = 47,955$), card risk drops 3.5-fold from a team's own defensive fifth (31.4%) to the attacking fifth (8.9%) and resets by -3.3 percentage points across half-time ($16.1\% \to 12.8\%, p = 4.1 \times 10^{-4}$), yet 98 Big-Five clubs compress into $[48.2, 58.0]$ in mean foul position ($r = -0.143$, inside a 2,000-draw matched spatial-Poisson null of $\bar{r} = -0.210, [-0.397, -0.015]$). Even restricting to late-game ($\geq 70'$), close-score (0 or +1) defensive/middle-third fouls (4,547 fouls, 25.7% carded) explains only 0.3% of between-club variance ($r = +0.053$). Fourth, across 2,528 consecutive club-season pairs, only 15.1% of single-season variance is true club signal ($r = +0.324$ vs. Poisson null maximum +0.056), so ignoring a club's one-season ratio in favour of the league mean lowers next-season prediction error by 11.6% (and Empirical-Bayes shrinkage by 13.3%), while all 31 historical bottom-1% four-match openings reverted to a rest-of-season median of 0.965.

1 Introduction

In professional association football, disciplinary statistics—fouls committed and yellow or red cards issued—are the most widely available behavioural records outside goals and shots. Because raw counts of fouls and bookings are published for almost every modern top-flight match, analysts, club officials, and supporters frequently construct a simple ratio: yellow cards per foul committed (or its reciprocal, fouls per card). When a dominant domestic club accumulates a low ratio of cards to fouls, the discrepancy is routinely interpreted either as institutional favouritism by match officials or as a deliberate tactical skill: committing fouls in pitch zones or match states where referees are reluctant to book [1, 17].

This paper demonstrates that reasoning directly from an aggregate cards-to-fouls ratio to a club-level behavioural or officiating claim commits a sequence of four methodological errors:

1. **Silent schema and denominator traps (Study 01):** Public football repositories exhibit severe longitudinal truncation, unrecorded rule differences in second-yellow accounting, and missing referee identifiers across 9 of 11 European leagues.
2. **Proportional misspecification and fixture-level confounding (Study 02):** Yellow cards are an affine rather than proportional function of fouls because conduct offences (dissent, delaying restarts, simula-

tion) enter as a positive intercept (13%–47% of cards). Moreover, conditioning on pre-match strength and testing the *opponents'* booking index proves that elevated card-per-foul rates are a property of high-quality fixtures for *both* teams on the pitch.

3. **Event-to-actor ecological compression (Study 03):** Although the spatial and temporal card gradient at the level of an individual foul is steep (31.4% in the defensive fifth vs. 8.9% in the attacking fifth; 25.7% in late-game close-score defensive/middle-third situations vs. 14.2% outside), clubs stand in nearly the same place on the pitch ([48.2, 58.0] mean x-coordinate), leaving 91.2% of between-club variation unexplained even on the most favourable tail statistic.
4. **Small-sample Poisson dominance (Study 04 & Sealed Protocol):** Across 2,528 consecutive club-season pairs, 84.9% of the cross-sectional variance in a single season's adjusted booking index is Poisson sampling noise. Consequently, a club's own one-season raw ratio is outperformed as an out-of-sample predictor of its next season by 11.6% simply by predicting the league mean, and by 13.3% via Empirical-Bayes shrinkage [5, 9].

Every empirical claim in this paper is bound by continuous-integration tests to deterministic JSON sidecars and Parquet snapshots committed in an open repository, accompanied by a six-attack adversarial pressure-test matrix (specification, aggregation, adjustment coarseness, prior work, baseline sufficiency, and cross-sectional unit count) and a sealed out-of-sample pre-registration tracking an extreme 2026–27 four-match opening (58 fouls, 1 yellow card).

2 Related Work

Referee bias, social pressure, and team strength. A rich econometric and sports-science literature examines officiating in football. Garicano et al. [10] and Dohmen [8] established that referees adjust injury time in response to score state and crowd pressure, while Boyko et al. [4] and Sors et al. [21] documented home-crowd effects on disciplinary sanctions that attenuated in spectator-free matches. Most relevant to our Study 02, Dawson et al. [7] analysed 2,660 English Premier League matches (1996–2003) using a bivariate Negative Binomial and Nash-equilibrium aggression model, finding significant referee-to-referee inconsistency alongside a team-strength gradient in total bookings. We extend Dawson et al. [7] to 58,013 matches across 11 leagues while conditioning explicitly on foul volume and decomposing own versus opponent indices within the same fixture.

Fouling incentives, spatial risk, and Simpson's paradox. Wright and Hirotsu [23] modelled when a professional foul is tactically rational, and Ridder et al. [18] estimated the scoring cost of red-card dismissals. Phatak et al. [17] compared fouls-per-card across five European leagues and demonstrated Simpson's paradox [20] when pooling across leagues with distinct officiating baselines. At the individual event level, Pappalardo et al. [15] released spatio-temporal event logs for the 2017–18 Big-Five European seasons, and Azmat and Yi [1] formulated an Expected Booking (xB) model of $P(\text{card} \mid \text{foul context})$. While Azmat and Yi [1] establish that event context predicts whether *a* foul is carded, our Study 03 tests whether real clubs actually separate along that gradient.

Reliability, shrinkage, and false discovery. Separating persistent skill from random noise in sports metrics relies on split-half reliability [6], Empirical-Bayes shrinkage [5, 9], partial pooling [12], and False Discovery Rate (FDR) control [3], alongside guards against regression-to-the-mean artifacts [2] and undisclosed researcher degrees of freedom [13, 16, 19]. Johnson et al. [14] raised the profile-stabilisation question for football match indicators; our Study 04 provides the exact variance decomposition and out-of-sample shrinkage validation for disciplinary indices across 11 leagues.

Table 1: Audit of 286 completed league-season files across 11 European top divisions (2000–01 to 2025–26, snapshot `2026-W35`; $N = 58,013$ matches with fouls and cards).

League	Fouls & Cards	Usable Files	Referee Column
England (`E0`)	2000–2025	26/26	2000–2025 (26)
Scotland (`SC0`)	2000–2025	26/26	25/26 (no 12–13)
Germany (`D1`)	00–01, 03–25	25/26	2000–2001 only (2)
Italy (`I1`)	2005–2025	21/26	2005–2006 only (2)
Spain (`SP1`)	2005–2025	21/26	Never (0)
France (`F1`)	2007–2025	19/26	Never (0)
Netherlands (`N1`)	2017–2025	9/26	Never (0)
Portugal (`P1`)	2017–2025	9/26	Never (0)
Turkey (`T1`)	2017–2025	9/26	Never (0)
Belgium (`B1`)	2019–2025	7/26	Never (0)
Greece (`G1`)	2019–2025	7/26	Never (0)
Total	**58,013 matches**	**179 / 286**	**55 / 286 files**

3 Study 01: Provenance and Structural Limits of Public Match Data

Before fitting any disciplinary model, we audited the primary open archive of match-level European football statistics (`football-data.co.uk`) across 11 top divisions—England (`E0`), Scotland (`SC0`), Germany (`D1`), Italy (`I1`), Spain (`SP1`), France (`F1`), Netherlands (`N1`), Belgium (`B1`), Portugal (`P1`), Turkey (`T1`), and Greece (`G1`)—over 26 completed seasons (2000–01 through 2025–26), comprising 286 league-season CSV files [11, 22].

Table 1 summarises four structural findings that constrain any valid multi-league disciplinary study:

1. **Silent absence of match statistics:** Only 179 of 286 league-season files (62.6%) contain foul and card columns (58,013 completed matches; 190 files when including the in-progress 2026–27 snapshot). The re-

maining 107 files download with HTTP 200 and parse cleanly into DataFrames containing only scores and betting odds. In Portugal, Netherlands, and Turkey, foul and card records begin only in 2017–18 (9 completed seasons); in Belgium and Greece, in 2019–20 (7 seasons).

2. **Interrupted series:** Germany carries fouls, cards, and referee identities in 2000–01 and 2001–02, drops all match statistics in 2002–03, and resumes fouls and cards (without referees) from 2003–04.
3. **Referee coverage contraction:** Match-official identity is recorded continuously only in England and Scotland (bar 2012–13), for two seasons each in Germany and Italy, and never in the other seven leagues.
4. **Schema widening without event expansion:** While the median file width more than quintupled from 25 columns in 2000–01 to 131 columns in 2025–26 due to bookmaker odds proliferation, columns describing pitch events rose from 0 to 12 by 2007 and have remained frozen for nineteen years. Moreover, two definitional traps prohibit cross-league pooling: English and Scottish feeds exclude the first yellow card of a second-bookable dismissal whereas continental feeds count both, and match-level yellow-card totals include non-foul bookings (dissent, time-wasting, bench sanctions).

4 Study 02: Deconfounding the Club Booking Index

Motivated by a viral public claim that FC Porto were systematically favoured by Portuguese referees based on a low raw cards-per-foul ratio, we pre-registered a test of whether any club in the Portuguese Primeira Liga (`P1`) exhibits an anomalous booking rate relative to the fouls it commits, and subjected the estimator to six adversarial pressure tests across all 11 European leagues (58,013 matches, 115,946 team-match observations).

4.1 The Naive Estimator and Its Four Failures

Let $Y_{i,m}$ and $F_{i,m}$ denote the yellow cards received and fouls committed by team i in match m. Under a naive proportional model $E_0[Y_{i,m}] = \bar{r}_{L,s} F_{i,m}$ (where $\bar{r}_{L,s} = \sum Y / \sum F$ within league L and season s), FC Porto's ten-season raw booking index $I_i^{\text{raw}} = \sum Y_{i,m} / \sum E_0[Y_{i,m}]$ is 0.890 and SL Benfica's is 0.885—both roughly 11% below parity (1.000). Four sequential diagnostics show why this raw index is an artifact.

Failure 1: Pre-match strength and match expectation. Using pre-match bookmaker odds (and Elo ratings where odds are absent) to compute each side's pre-match strength differential $s_{i,m}$, Figure 1(a) partitions all 115,946 team-match rows into five expectation bands. In Portugal, a heavy underdog is booked at 1.121× its foul-implied expectation whereas a heavy favourite is booked at 0.854×—a 31.3% swing with no club identity in the model. Across all 11 European leagues, the heavy-underdog band exceeds the heavy-favourite band in 11 of 11 competitions (median 1.098 vs. 0.856, strictly monotone across all five bands in 7 of 11 leagues). Because domestic giants are favourites in most fixtures, attributing the favourite discount to club-specific referee bias confounds club identity with match situation.

Failure 2: Affine misspecification and non-foul intercepts. The proportional model assumes zero cards at zero fouls ($a = 0$). Fitting the affine specification $E[Y_{i,m} \mid F_{i,m}] = a_{L,s} + b_{L,s} F_{i,m}$ within each league-season reveals a positive intercept $a_{L,s}$ accounting for 13% to 47% of mean yellow cards across the 11 leagues (reaching nearly half of all bookings in Greece and Italy). Figure 2(a) shows the empirical consequence: observed yellows per foul $Y/F = a/F + b$ declines monotonically with foul volume across all nine bins, from 0.197 when a side commits ≤ 6 fouls (5.1 bin mean) to 0.122 when it commits > 20 fouls (23.4 bin mean). Cross-checking against the Wyscout event-coded sub-types in our Study 03 dataset (`typology.json`) confirms the physical mechanism: four non-challenge conduct labels (*Protest*, *Time lost foul*, *Simulation*, and *Out of game foul*) produce 642 of the 7,346 total cards (8.74% of all cards, carded at 99.1%), with individual club non-play card shares ranging from 1.5% to 17.9% (median 8.1%, split-half reliability $r = 0.251$, Spearman–Brown corrected 0.402). Forcing the expectation line through the origin mechanically inflates low-foul dominant teams and deflates high-foul teams.

Failure 3: The opponent test (fixture quality vs. club trait). Once the affine expectation is adjusted continuously for pre-match strength, a club's long-run strength still correlates positively with its own adjusted booking index across the 246 clubs in the 11 leagues ($r = +0.484, p = 7.8 \times 10^{-16}$; Spearman $\rho = +0.349$). However, computing the exact same adjusted booking index for each club's *opponents* in the matches they play against that club yields $r = +0.283$ ($p = 6.6 \times 10^{-6}$), while a club's own index and its opponents' index are nearly independent ($r = +0.116, p = 0.069$). Even more decisively, holding club identity fixed and partitioning each club's own matches into terciles of opponent quality shows that a team's booking index rises with the strength of the opponent it faces in 11 of 11 leagues: across the 11-league median, a club is booked at 0.938 against weakest-tercile opponents, 0.993 against middle-tercile opponents, and 1.088 against strongest-tercile opponents ($0.939 \rightarrow 0.973 \rightarrow 1.092$ in Portugal). Thus, yellows per foul scale with the quality of the *fixture* for both teams. Testing whether the residual belongs to the head coach across 42 multi-club managerial spell pairs in Portugal yields a null correlation of $r = -0.076$ (5,000-permutation $p = 0.628$).

Failure 4: Ecological reversal across aggregation levels. Across the 11 league averages, the steepness of a league's strength gradient correlates with its

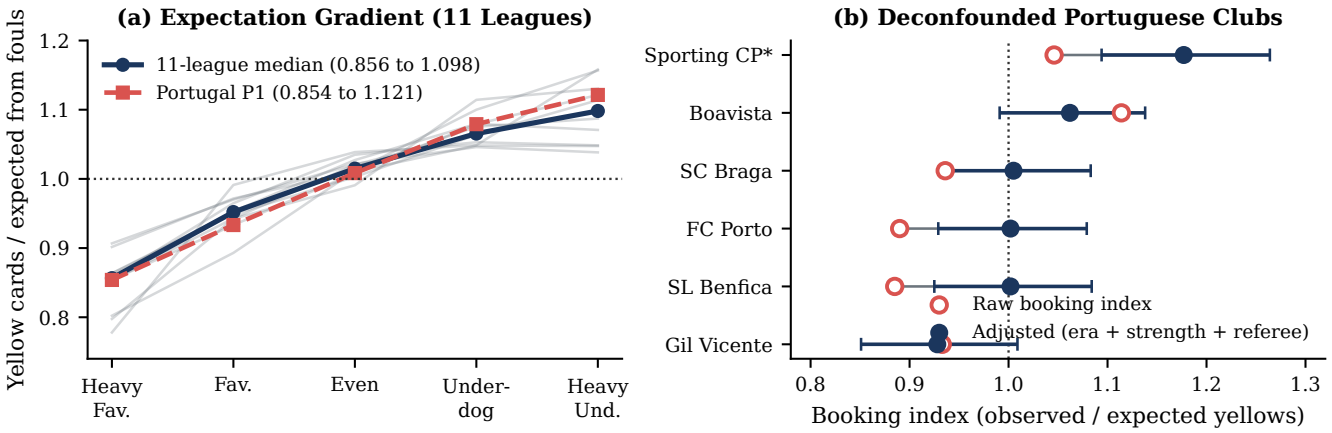


Figure 1: Study 02: Match-expectation gradient and club-level deconfounding. **(a)** Ratio of observed yellow cards to fouls-implied expectation across five pre-match win-probability bands in 11 European leagues ($N = 58,013$ matches); heavy underdogs are booked at 1.098× median expectation vs. 0.856× for heavy favourites (1.121 vs. 0.854 in Portugal, a 31.3% swing). **(b)** Raw vs. era-, situation-, and leave-one-club-out referee-adjusted booking indices across Portuguese clubs ($N = 2,737$ official appointments, 95% CIs). FC Porto (0.890 → 1.002) and SL Benfica (0.885 → 1.002) collapse to the league average; only Sporting CP (1.177, BH $p = 0.0004$) separates.

mean home goal difference at $r = +0.782$ ($p = 0.0045$). Yet when the same panel is disaggregated into 62 three-season blocks, Figure 2(b) shows the correlation dropping to $r = +0.238$ ($p = 0.063$) when pooled and reversing to $r = -0.086$ ($p = 0.504$, negative in 5 of 6 leagues with ≥ 4 blocks) within leagues over time, despite home advantage falling from 0.349 goals/match in 2001 to 0.220 in 2025.

4.2 Leave-One-Club-Out Referee Adjustment in Portugal

Because `football-data.co.uk` omits Portuguese referees, we harvested 2,737 official appointments across ten seasons and joined 98.4% of them to our match records (5,508 team-matches) by date and scoreline validation. Among the 32 Portuguese officials with ≥ 40 matches, individual referee booking multipliers span 0.755 to 1.324—as wide as in England—but referee assignment across the 26 ranked clubs (≥ 80 matches) is tightly balanced, with every club's mean referee draw lying between 0.971 and 1.021.

Estimating each referee's strictness multiplier $\hat{\rho}_{r,-i}$ *excluding* all matches involving club i (preventing a club's own card rate from contaminating its official baseline), Figure 1(b) and Table 2 report the fully adjusted club booking indices:

- **FC Porto:** Raw 0.890 → situation-adjusted 1.018 → referee-adjusted **1.002** (95% CI [0.929, 1.079], two-sided $p = 0.964$, BH-adjusted $p = 0.965$).
- **SL Benfica:** Raw 0.885 → situation-adjusted 1.009 → referee-adjusted **1.002** (95% CI [0.925, 1.084], raw $p = 0.965$, BH-adjusted $p = 0.965$).
- **Sporting CP:** Raw 1.046 → situation-adjusted 1.179 → referee-adjusted **1.177** (95% CI [1.094, 1.264], raw $p = 1.6 \times 10^{-5}$, BH-adjusted $p = 0.0004$, Bonferroni $p = 0.0004$).

Table 2: Deconfounded booking index for selected Portuguese clubs across ten seasons ($N = 5,508$ team-matches, 26 clubs with ≥ 80 matches, 2,737 referee appointments).

Club	Raw	Sit. Adj.	Ref. Adj.	95% CI	Raw p	BH p
Sporting CP	1.046	1.179	**1.177**	[1.094, 1.264]	0.00002	**0.0004**
Boavista	1.114	1.067	1.062	[0.991, 1.138]	0.090	0.635
SC Braga	0.936	1.020	1.005	[0.931, 1.083]	0.914	0.965
FC Porto	0.890	1.018	**1.002**	[0.929, 1.079]	0.964	0.965
SL Benfica	0.885	1.009	**1.002**	[0.925, 1.084]	0.965	0.965
Gil Vicente	0.933	0.920	0.928	[0.851, 1.009]	0.081	0.635

Under Benjamini–Hochberg FDR control at $q = 0.10$ across all 26 clubs, only Sporting CP separates; the runner-up (Gil Vicente, 0.928, raw $p = 0.081$, BH $p = 0.635$) and third-ranked club (Boavista, 1.062, raw $p = 0.090$, BH $p = 0.635$) are indistinguishable from sampling variation.

5 Study 03: The Spatial and Temporal Lever Nobody Pulls

Could a club achieve a genuinely low card-per-foul rate by committing its fouls higher up the pitch or at safer moments? To test this mechanism at the event level, we analysed all 47,955 fouls across 1,826 matches and 98 clubs (≥ 200 fouls each) in the 2017–18 English, French, German, Italian, and Spanish top divisions from Pappalardo et al. [15] (overall base card rate 15.32%).

5.1 Steep Event Gradients and the Half-Time Discontinuity

At the level of an individual foul, pitch location, elapsed time, and score state exert massive effects on card prob-

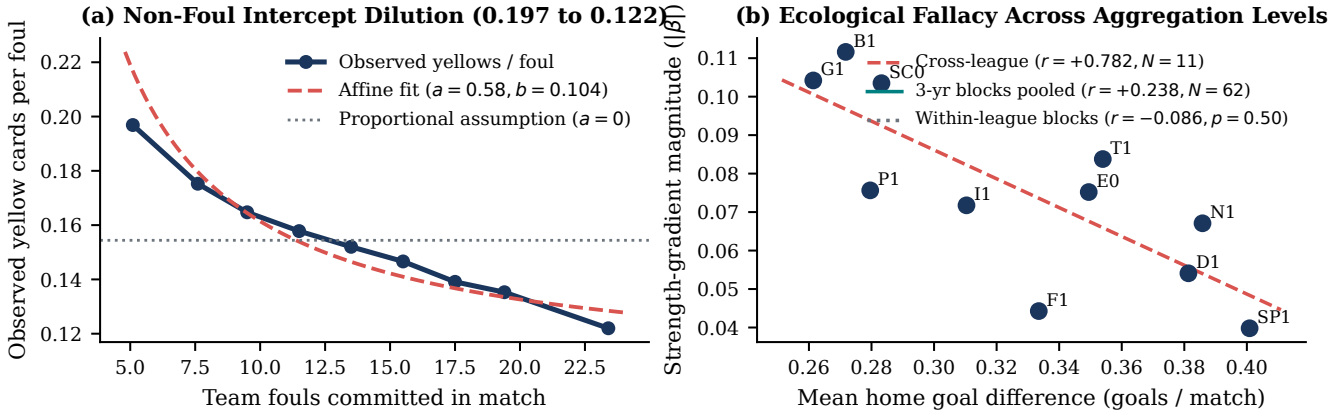


Figure 2: Study 02: Non-foul intercept dilution and multi-level ecological reversal. **(a)** Observed yellow cards per foul across team foul-count bins ($N = 115,946$ team-matches), declining monotonically from 0.197 at ≤ 6 fouls to 0.122 at > 20 fouls because non-foul conduct cards enter as a positive affine intercept $a > 0$ (13%–47% of mean cards). **(b)** Cross-league correlation between strength-gradient magnitude and home goal difference ($r = +0.782, N = 11$) vanishes across pooled three-season blocks ($r = +0.238, N = 62$) and reverses sign within leagues over time ($r = -0.086, p = 0.504$, negative in 5 of 6 leagues).

ability (Figure 3):

- **Longitudinal and lateral location:** A foul in the fouling team's own defensive fifth ($x \leq 20$) draws a card **31.4**% of the time, compared with **8.9**% in the attacking fifth ($x > 80$)—a 3.54× ratio (ranging from 2.47× in England to 4.97× in Germany). By pitch thirds, card risk is 24.1% in the defensive third, 14.0% in the middle third, and 10.6% in the attacking third (22.8%, 12.7%, and 9.3% when non-play conduct cards are removed). Central fouls ($|y - 50| \leq 10$) are carded at 19.0% versus 13.0% wide ($|y - 50| > 40$).
- **Elapsed time and the half-time reset:** Card probability quadruples from **6.5**% in the opening quarter-hour (0–15′) to **25.6**% in second-half stoppage time ($> 90'$). Crucially, the trajectory is not monotonic: card rate climbs to **16.1**% in the final five minutes of the first half (40–45′), then drops by **3.3** percentage points to **12.8**% in the opening five minutes of the second half (45–50′, χ^2 contingency $p = 4.1 \times 10^{-4}$). Because physiological fatigue does not discharge to first-half levels across a fifteen-minute interval, this discontinuity reflects within-half referee tolerance accumulation and reset.
- **Score state:** Fouls committed while trailing are carded at **18.1**%, while leading at 16.2%, and while level at **13.3**%.

5.2 Why the Event Gradient Vanishes Between Clubs

Despite a 3.5-fold spatial gradient across the 100-unit pitch, the season-average foul position across all 98 Big-Five clubs spans only **48.2** to **58.0**—less than one-tenth of the pitch (Figure 3(a)). Using Cronbach [6] split-variance reliability to net out within-club sampling error, 70.1% of the between-club variance in mean foul x-position is genuine tactical variation (versus only 13.7% for mean foul minute), and shifting from the worst to the best observed spatial-temporal foul mix among the 98 clubs is worth 17.9% of the base card rate (16.6% vs. 13.9%, and at most a 7.3% reduction below a league's base rate in the most generous league, Spain).

Across the 98 clubs, mean foul x-coordinate correlates with club card rate at $r = -0.143$ (95% CI $[-0.332, +0.057], p = 0.160$; 80% power threshold $|r| = 0.28$), and an additive logistic regression in $(x, x^2, |y - 50|, \text{minute}, \text{score state})$ explains only **3.1**% of the between-club standard deviation in cards per foul.

Matched spatial-Poisson and permutation nulls. To separate "no club exploits the spatial gradient" from "the observed [48.2, 58.0] placement range is too narrow to overcome single-season Poisson noise," we executed two 2,000-draw Monte Carlo nulls. Under a spatial-permutation null that randomly reassigns club mean x-positions across the 98 clubs, $\bar{r} = +0.002$ (95% interval $[-0.193, +0.199]$, two-sided $p = 0.161$). Under a matched spatial-Poisson null where each club's actual foul coordinates in [48.2, 58.0] are held fixed and club card totals are drawn from Poisson$(\sum_{j \in c} \hat{p}_{\text{spatial}}(x_j, y_j))$, the expected correlation is $\bar{r} = \mathbf{-0.210}$ with a 95% sampling interval of $[\mathbf{-0.397}, \mathbf{-0.015}]$. The observed $r = -0.143$ sits comfortably inside *both* distributions: because clubs compress into a 9.8-unit band near midfield, the mechanical slope induced by the spatial curve is smaller than the Poisson sampling noise of a 38-match season.

Tail-share versus volume proxy. Replacing mean foul x with a tail statistic—the share of a club's fouls committed in its own defensive third ($x < 33.3$, spanning 16.1% to 29.9% across the 98 clubs)—strengthens the club-level correlation to $r = \mathbf{+0.296}$ (95% CI $[+0.103, +0.467]$, $R^2 = \mathbf{8.8}$%; $r = +0.305, R^2 = 9.3$%

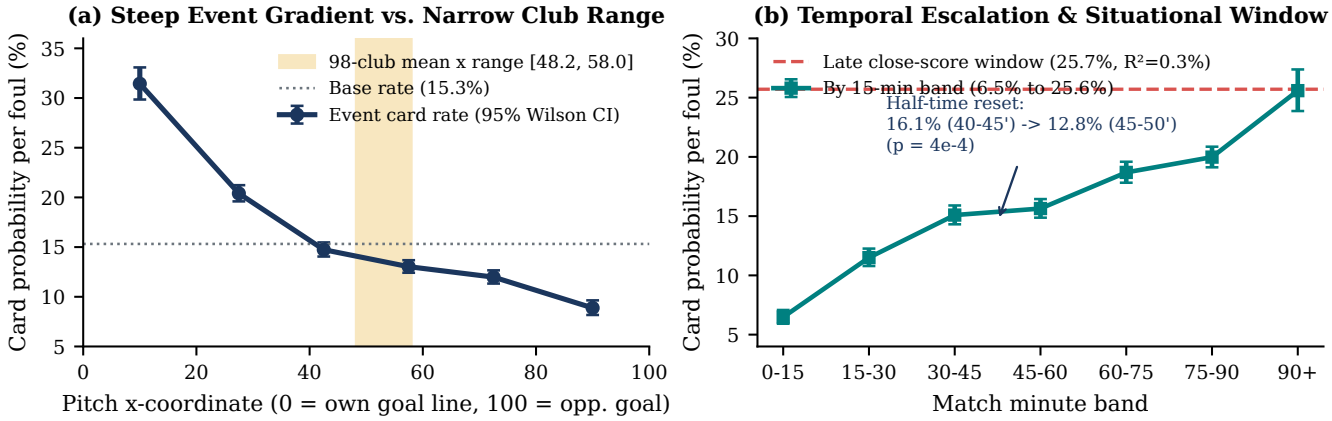


Figure 3: Study 03: Event-level disciplinary gradients vs. club-level spatial compression ($N = 47,955$ **fouls,** $1,826$ **matches,** 98 **Big-Five clubs). (a)** Card probability per foul falls 3.5-fold from a team's own defensive fifth (31.4%) to the attacking fifth (8.9%), but all 98 clubs' season-mean foul x-coordinates compress into $[48.2, 58.0]$ (amber band), yielding $r = -0.143$ ($[-0.332, +0.057]$), indistinguishable from a $2,000$-draw matched spatial-Poisson null ($\bar{r} = -0.210$, $[-0.397, -0.015]$). **(b)** Temporal escalation from 6.5% (0–15′) to 25.6% (90+′), featuring a significant -3.3 pp half-time reset (16.1% $\rightarrow$ 12.8%, $p = 4.1\times10^{-4}$) and a 25.7% rate inside the late-game ($\geq 70'$), close-score $(0, +1)$ defensive/middle-third tactical window ($R^2 = 0.3\%$ between clubs).

on play-only fouls). Conversely, expressing defensive-third fouls as a per-match count (1.76 to 4.32, median 3.07) correlates with cards per match at $r = +0.670$ ($[+0.544, +0.766]$) solely because total fouls per match correlates with cards per match at $r = +0.696$ ($[+0.577, +0.786]$).

Situational tactical-fouling interaction test. A natural objection is that season averages mask cynical fouls committed in high-leverage moments: late in the match (minute $\geq$ 70), in a close game (score_diff $\in \{0, +1\}$), in the defensive or middle third ($x < 66.7$). Isolating this exact window captures **4, 547** fouls (9.5% of all fouls), where the card rate jumps to **25.7**% (**26.8**% across the $2,069$ fouls committed while protecting a one-goal lead) versus **14.2**% outside the window. Within these $4,547$ situational fouls, a club's mean foul x-position correlates significantly with its situational card rate at $r = -\mathbf{0.268}$ (95% CI $[-0.443, -0.074]$, $p = 0.0076$). However, because clubs commit a median of only 45.5 such fouls per season (22.4% variance reliability) and their situational foul share spans only 4.5% to 14.8%, club situational share correlates with full-season card rate at $r = +\mathbf{0.053}$ ($[-0.147, +0.249]$, $R^2 = \mathbf{0.3}\%$), and augmenting the logistic model with full position $\times$ minute $\times$ score-state interaction terms explains only 2.5% of the between-club SD ($R^2 = 4.2\%$).

6 Study 04 & Sealed Protocol: Variance Decomposition, Shrinkage, and Early-Season Reversion

To determine how much of a single season's adjusted booking index reflects a repeatable club trait versus Poisson sampling noise, we constructed $2,528$ consecutive same-division club-season pairs ($\geq$ 25 matches per season) across all 11 leagues in our $58,013$-match archive.

6.1 Persistence, Poisson Null, and Variance Decomposition

Across the $2,528$ consecutive pairs, a club's situation- and era-adjusted booking index in season t correlates with its index in season $t + 1$ at $r = +\mathbf{0.324}$ ($R^2 = 10.5\%$, $p = 6.5\times10^{-63}$), and is positive in **11** of **11** leagues (ranging from +0.172 in Scotland to +0.427 in Italy; Figure 4(a)). Resampling each club-season's yellow cards $2,000$ times from Poisson(expected$_{i,s}$) while holding fixture lists and expectations fixed produces a matched null with $\bar{r} = -0.0004$, 95th percentile +0.032, and a maximum across all $2,000$ draws of $+\mathbf{0.056}$ ($p < 0.0005$).

Decomposing the observed cross-sectional variance $\sigma^2_{\text{obs}} = 0.01735$ into expected Poisson sampling variance $\sigma^2_{\text{Poisson}} = E[Y/E[Y]^2] = 0.01473$ and true club heterogeneity $\sigma^2_{\text{true}} = \sigma^2_{\text{obs}} - \sigma^2_{\text{Poisson}} = 0.00262$ reveals that only **15.1**% of the spread in single-season booking indices is a true club property ($\sigma_{\text{true}} = \mathbf{0.051}$), while **84.9**% is the random Poisson noise of one season's cards.

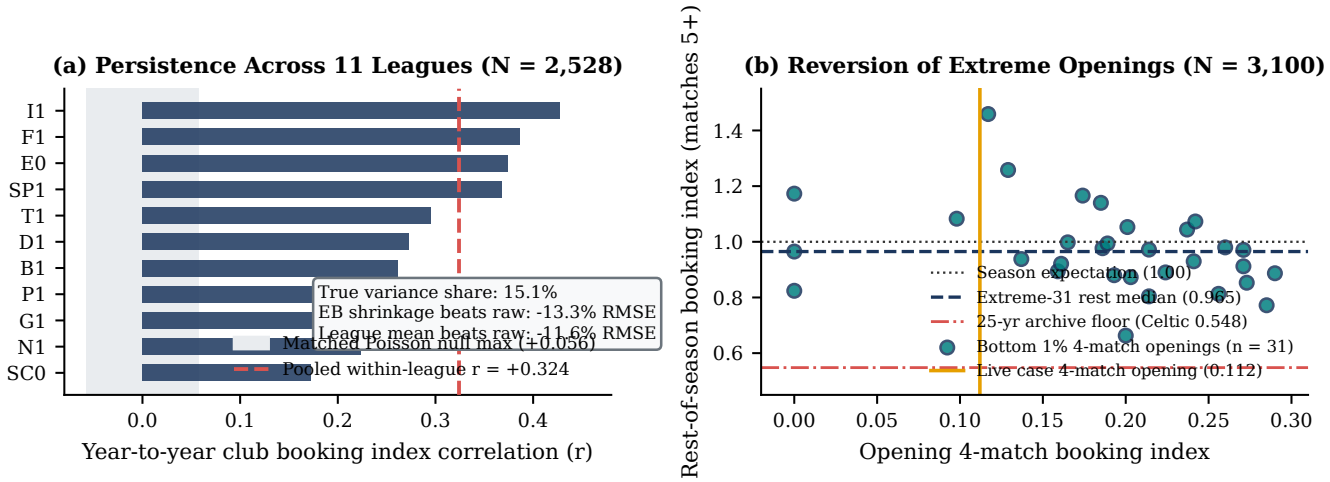


Figure 4: Study 04 & Sealed Out-of-Sample Protocol: Persistence, Empirical-Bayes shrinkage, and reversion of extreme openings. (a) Year-to-year club booking index persistence across 2,528 consecutive club-season pairs in 11 leagues ($r = +0.324, p = 6.5 \times 10^{-63}$, positive in 11/11 leagues vs. 2,000-draw matched Poisson null maximum +0.056); only 15.1% of single-season variance is true club signal ($\sigma_{\text{true}} = 0.051$). **(b)** Across 3,100 completed team-seasons, all 31 bottom-1% four-match openings (index ≤ 0.292) reverted over the remainder of the season to a median index of 0.965 and a minimum of 0.663 (above the 25-year archive floor of 0.548).

6.2 Out-of-Sample Shrinkage Validation

Because single-season reliability is 0.151, ranking clubs on one season of unshrunk ratios is dominated by estimation error. Across 2,432 consecutive club-season pairs (≥ 30 matches), we evaluated three forecasts of a club's season-$t+1$ booking rate: (i) the club's own season-t raw ratio, (ii) the league mean (ignoring the club entirely), and (iii) an Empirical-Bayes Beta-Binomial shrunken estimator $\hat{\theta}_i = \frac{n_i}{n_i+n_0} r_i + \frac{n_0}{n_i+n_0} \bar{r}_L$ [5, 9]. Out-of-sample RMSE is 0.0267 for the club's raw ratio, 0.0236 for the league mean, and 0.0232 for the shrunken estimate: **ignoring the club and predicting the league mean beats the club's own one-season ratio by** 11.6%, and **Empirical-Bayes shrinkage beats the raw ratio by** 13.3%.

6.3 Sealed Four-Match Protocol and Historical Reversion

At the start of the 2026–27 Portuguese season (snapshot `2026-W36`, 33 matches through matchweek 4), FC Porto committed 58 fouls and received 1 yellow card across 4 matches. Under our frozen pre-registered affine model ($1.1259 + 0.095491 \times$ fouls) and heavy-favourite multiplier (0.830), expected cards fell from 10.04 (era-only) to 8.34, yielding an adjusted index of **0.120** (95% CI [0.003, 0.668]), the sole club among 18 surviving a Benjamini–Hochberg screen at FDR = 0.10 ($p_{\text{BH}} = 0.081$ vs. 0.017 without the favourite adjustment). Applying the most generous spatial placement ceiling observed across the five European event-data leagues (Spain's 7.3% reduction) moves expected cards only from 8.34 to 7.73 (index 0.129, $p = 0.0077$), accounting for just **1.1**% of the 88.0% shortfall.

To establish the historical base rate without waiting for season completion, we severed all 3,100 completed team-seasons in our 11-league archive after match 4 (Figure 4(b)). Across all 3,100 team-seasons, the four-match opening index correlates with the rest-of-season index (matches 5+) at only $r = +\mathbf{0.118}$. Exactly 4 historical openings were at least as extreme as Porto's unadjusted opening (0.112), and the most lenient 1% (≤ 0.292, 31 team-seasons) finished with a rest-of-season median index of **0.965** and a minimum of **0.663**—none approaching the 25-year full-season archive floor (Celtic 2024–25 at 0.548). On 30 August 2026, we sealed a prospective decision rule on Porto's yellow-card total over the subsequent seven matchweeks (matchweeks 4–10, resolving 15 November 2026): ≤ 7 yellows classifies the run as persistent (22.6% historical base rate among the 31 extreme openings vs. 0.62% in Portugal, where the median seven-match stretch is 19 cards and Porto's nine-season historical minimum is 14), 8–11 as ambiguous (29.0%), and ≥ 12 as noise (48.4%).

7 Conclusion

Across 58,013 matches and 47,955 spatio-temporally coded fouls, apparent club-level disciplinary anomalies dissolve when subjected to proper statistical controls. Raw cards-per-foul ratios confound club identity with match-winner expectation (31.3% swing), non-foul conduct intercepts (13%–47% of bookings), and fixture-level intensity shared with the opponent ($r = +0.283$). At the event level, the 3.5-fold spatial card gradient goes largely unexploited across clubs because team foul distributions

compress into a narrow midfield band ($[48.2, 58.0]$), and at the season level, 84.9% of cross-sectional variance is Poisson noise, making Empirical-Bayes shrinkage mandatory before any disciplinary ranking is interpreted.

References

[1] Amod Azmat and Sun Sheng Yi. Investigating fouling efficiency in football using expected booking (xb) model. *arXiv preprint*, 2024. doi: 10.48550/arXiv.2401.08718. URL https://arxiv.org/abs/2401.08718.

[2] Adrian G. Barnett, Jolieke C. van der Pols, and Annette J. Dobson. Regression to the mean: What it is and how to deal with it. *International Journal of Epidemiology*, 34(1):215–220, 2005. doi: 10.1093/ije/dyh299.

[3] Yoav Benjamini and Yosef Hochberg. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B*, 57(1):289–300, 1995. doi: 10.1111/j.2517-6161.1995.tb02031.x.

[4] Ryan H. Boyko, Adam R. Boyko, and Mark G. Boyko. Referee bias contributes to home advantage in english premiership football. *Journal of Sports Sciences*, 25(11):1185–1194, 2007. doi: 10.1080/02640410601038576.

[5] Lawrence D. Brown. In-season prediction of batting averages: A field test of empirical bayes and bayes methodologies. *The Annals of Applied Statistics*, 2(1):113–152, 2008. doi: 10.1214/07-AOAS138.

[6] Lee J. Cronbach. Coefficient alpha and the internal structure of tests. *Psychometrika*, 16:297–334, 1951. doi: 10.1007/BF02310555.

[7] Peter Dawson, Stephen Dobson, John Goddard, and John Wilson. Are football referees really biased and inconsistent? evidence on the incidence of disciplinary sanction in the english premier league. *Journal of the Royal Statistical Society: Series A*, 170(1):231–250, 2007. doi: 10.1111/j.1467-985X.2006.00451.x.

[8] Thomas J. Dohmen. The influence of social forces: Evidence from the behavior of football referees. *Economic Inquiry*, 46(3):411–424, 2008. doi: 10.1111/j.1465-7295.2007.00112.x.

[9] Bradley Efron and Carl Morris. Data analysis using stein's estimator and its generalizations. *Journal of the American Statistical Association*, 70(350):311–319, 1975. doi: 10.1080/01621459.1975.10479864.

[10] Luis Garicano, Ignacio Palacios-Huerta, and Canice Prendergast. Favoritism under social pressure. *Review of Economics and Statistics*, 87(2):208–216, 2005. doi: 10.1162/0034653053970267.

[11] Timnit Gebru, Jamie Morgenstern, Briana Vecchione, Jennifer Wortman Vaughan, Hanna Wallach, Hal Daumé III, and Kate Crawford. Datasheets for datasets. *Communications of the ACM*, 64(12):86–92, 2021. doi: 10.1145/3458723.

[12] Andrew Gelman, Jennifer Hill, and Masanao Yajima. Why we (usually) don't have to worry about multiple comparisons. *Journal of Research on Educational Effectiveness*, 5(2):189–211, 2012. doi: 10.1080/19345747.2011.618213. URL https://arxiv.org/abs/0907.2478.

[13] John P. A. Ioannidis. Why most published research findings are false. *PLoS Medicine*, 2(8):e124, 2005. doi: 10.1371/journal.pmed.0020124.

[14] K. Johnson, A. Murphy, and R. Bower. Notational analysis: Establishing stable performance profiles in a-league football. *Journal of Science and Medicine in Sport*, 12 (S2):e118–e119, 2010. doi: 10.1016/j.jsams.2009.10.244.

[15] Luca Pappalardo, Paolo Cintia, Alessio Rossi, Emanuele Massucco, Paolo Ferragina, Dino Pedreschi, and Fosca Giannotti. A public data set of spatio-temporal match events in soccer competitions. *Scientific Data*, 6(1):236, 2019. doi: 10.1038/s41597-019-0247-7.

[16] Roger D. Peng. Reproducible research in computational science. *Science*, 334(6060):1226–1227, 2011. doi: 10.1126/science.1213847.

[17] Ashwin Phatak, Robert Rein, and Daniel Memmert. The dirty league: English premier league provides higher incentives for fouling as compared to other european soccer leagues. *Journal of Human Kinetics*, 80:263–276, 2021. doi: 10.2478/hukin-2021-0095. URL https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8607766/.

[18] Geert Ridder, J. S. Cramer, and Peter Hopstaken. Down to ten: Estimating the effect of a red card in soccer. *Journal of the American Statistical Association*, 89(427):1124–1127, 1994. doi: 10.1080/01621459.1994.10476850.

[19] Joseph P. Simmons, Leif D. Nelson, and Uri Simonsohn. False-positive psychology: Undisclosed flexibility in data collection and analysis allows presenting anything as significant. *Psychological Science*, 22(11):1359–1366, 2011. doi: 10.1177/0956797611417632.

[20] E. H. Simpson. The interpretation of interaction in contingency tables. *Journal of the Royal Statistical Society: Series B*, 13(2):238–241, 1951. doi: 10.1111/j.2517-6161.1951.tb00088.x.

[21] Fabrizio Sors, Massimo Grassi, Tiziano Agostini, and Mauro Murgia. The sound of silence in association football: Home advantage and referee bias decrease in matches played without spectators. *European Journal of Sport Science*, 21(12):1597–1605, 2021. doi: 10.1080/17461391.2020.1845814.

[22] Mark D. Wilkinson et al. The fair guiding principles for scientific data management and stewardship. *Scientific Data*, 3:160018, 2016. doi: 10.1038/sdata.2016.18.

[23] Mike Wright and Nobuyoshi Hirotsu. The professional foul in football: Tactics and deterrents. *Journal of the Operational Research Society*, 54(3):213–221, 2003. doi: 10.1057/palgrave.jors.2601506.